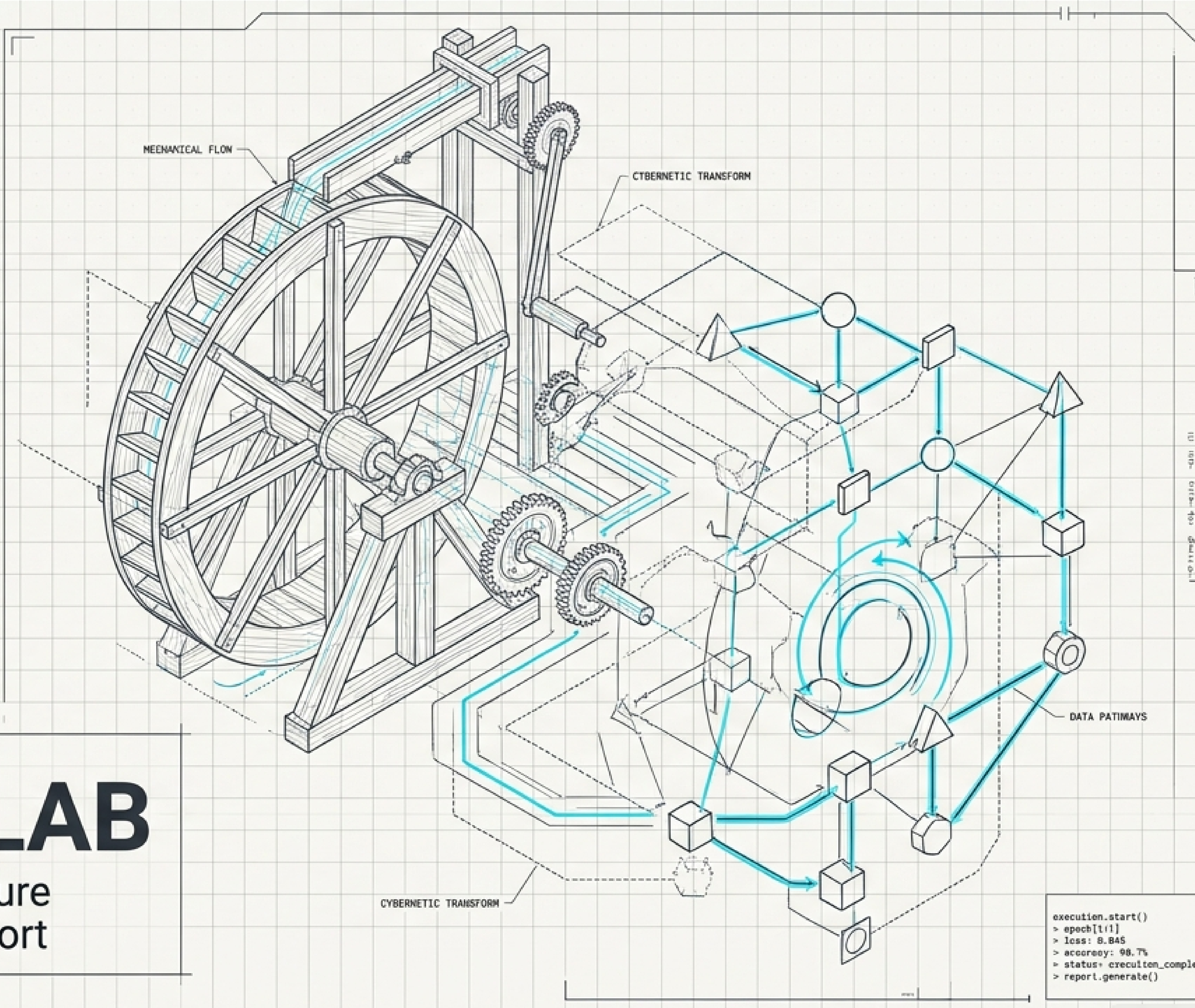
```
sys.boot(kernel='M_generator')
> init_eonget: 16.8
> config.load('avolutionary_search.yaml')
> agent.init(onb4)
> ccator: londy
```

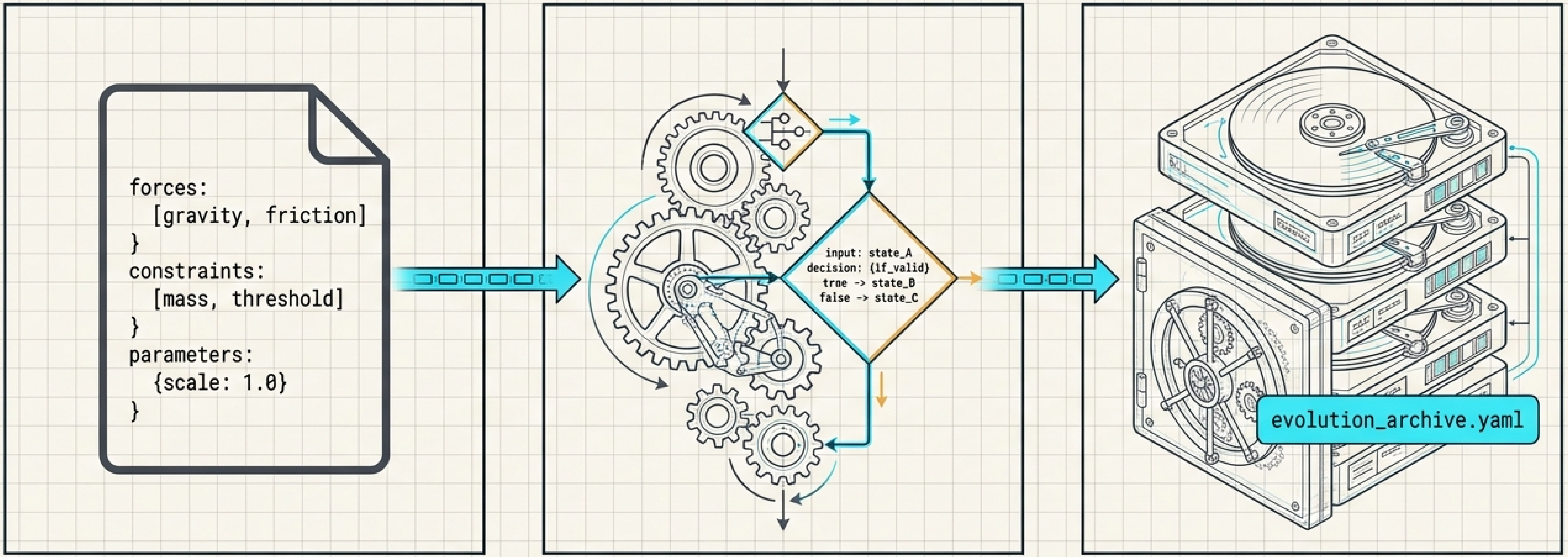


LEONARDO LAB

Evolutionary Search Architecture & Generation-1 Execution Report

```
execution.start()
> epoch[1/1]
> loss: 0.045
> accuracy: 98.7%
> status: execution_complete
> report.generate()
```

The architecture rests on three pillars ensuring cumulative mechanical adaptation



Pillar 1: Initial Genome

Parsed YAML ontology extracting A4 generator-level mechanics from Leonardo da Vinci's notebooks. Emphasizes physical forces over domain narratives.

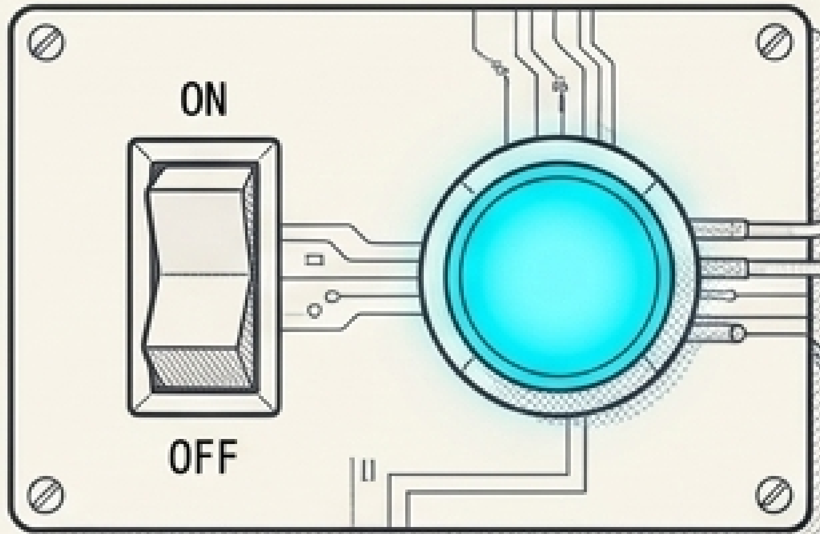
Pillar 2: GASO Engine

A rigid mechanical state machine driving sequential execution and managing evolutionary constraints.

Pillar 3: Evolution Archive

The phenotypic memory enabling cumulative adaptation.

Execution is governed by strict mechanical constraints and budgeted resources



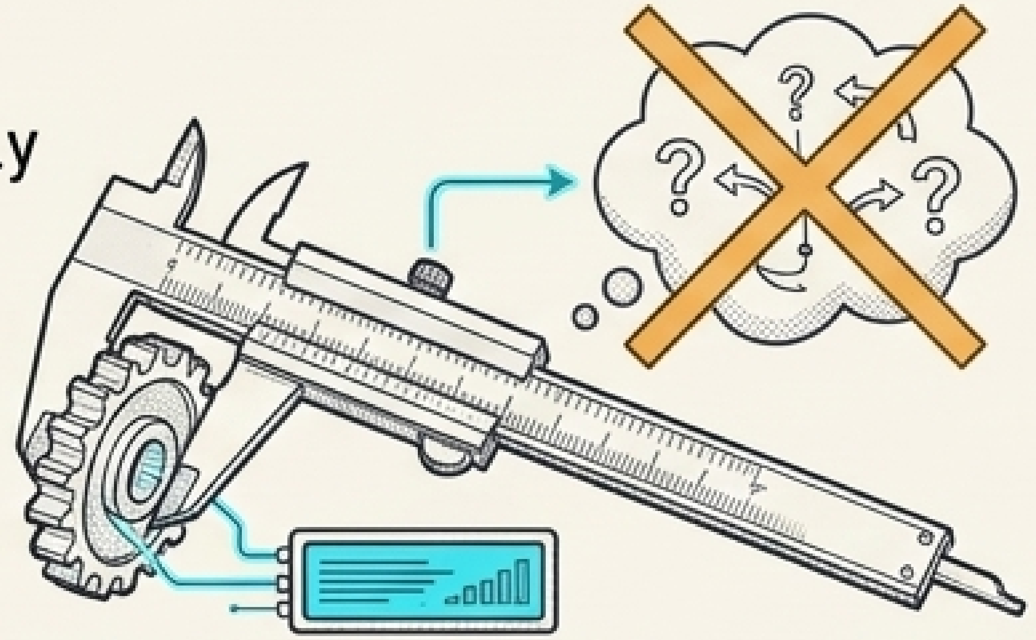
Fitness First

Optimization cannot begin without an explicit objective and search budget.

Parameters: [mortaml
{scale: scal: 10}
parasdout => -1

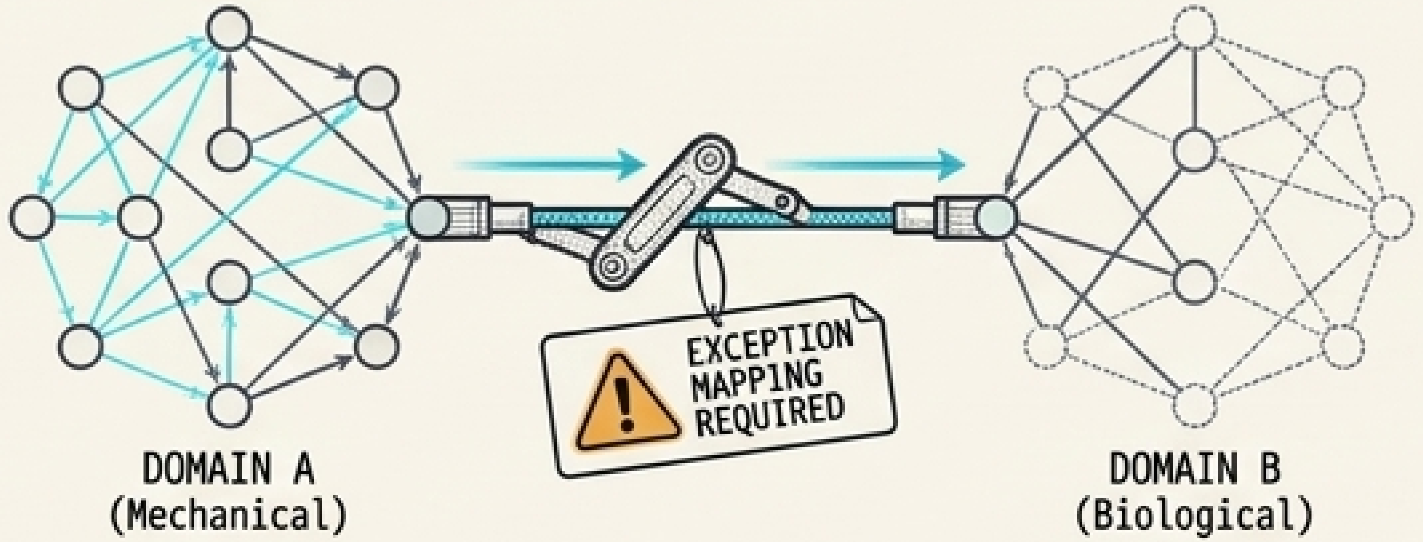
Mechanical Mutations Only

Mutations apply exclusively to extracted mechanical degrees of freedom, strictly rejecting abstract or semantic modifications.



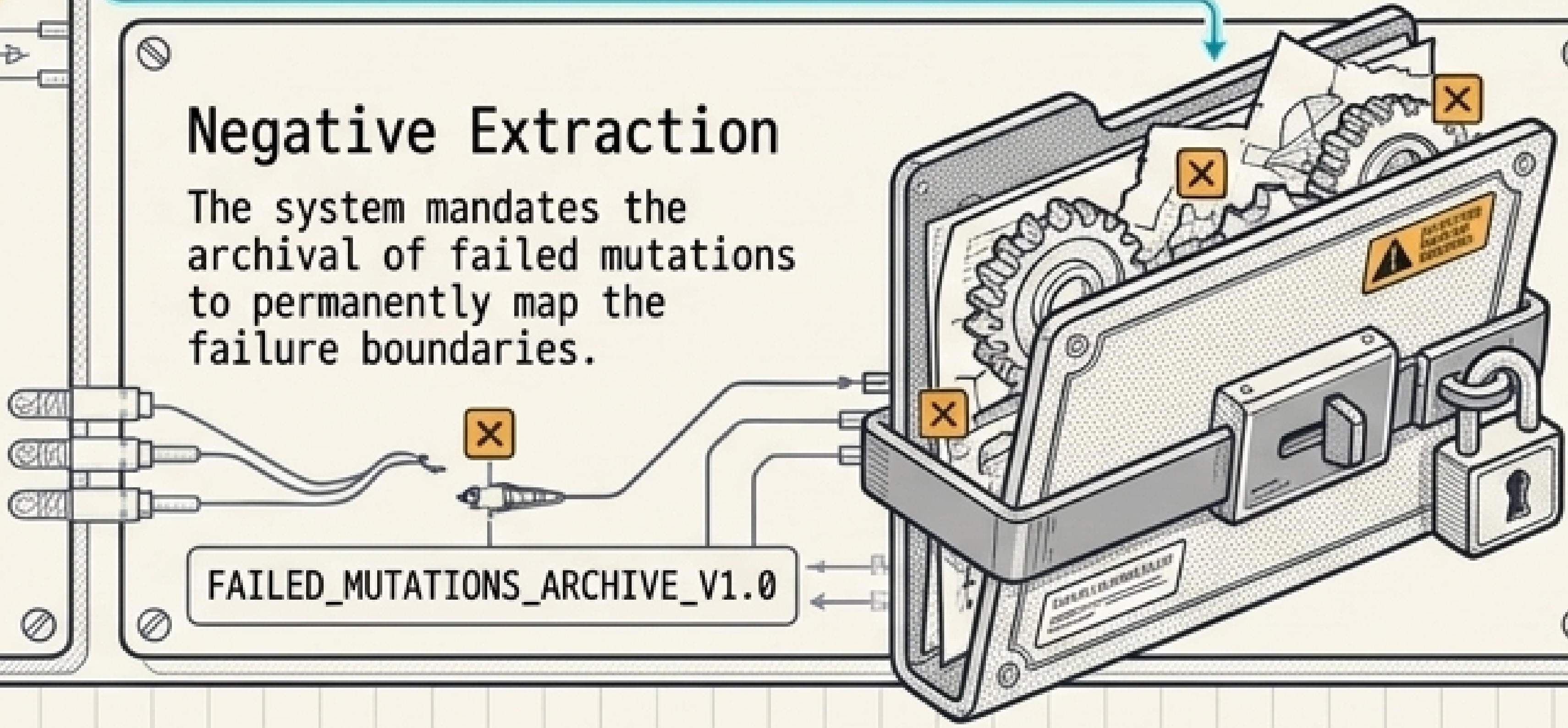
Role-Preserving Transfer

Structural transfers must explicitly map domain exceptions (e.g., noting ‘inventory storage’ has no biological blood-circulation equivalent).

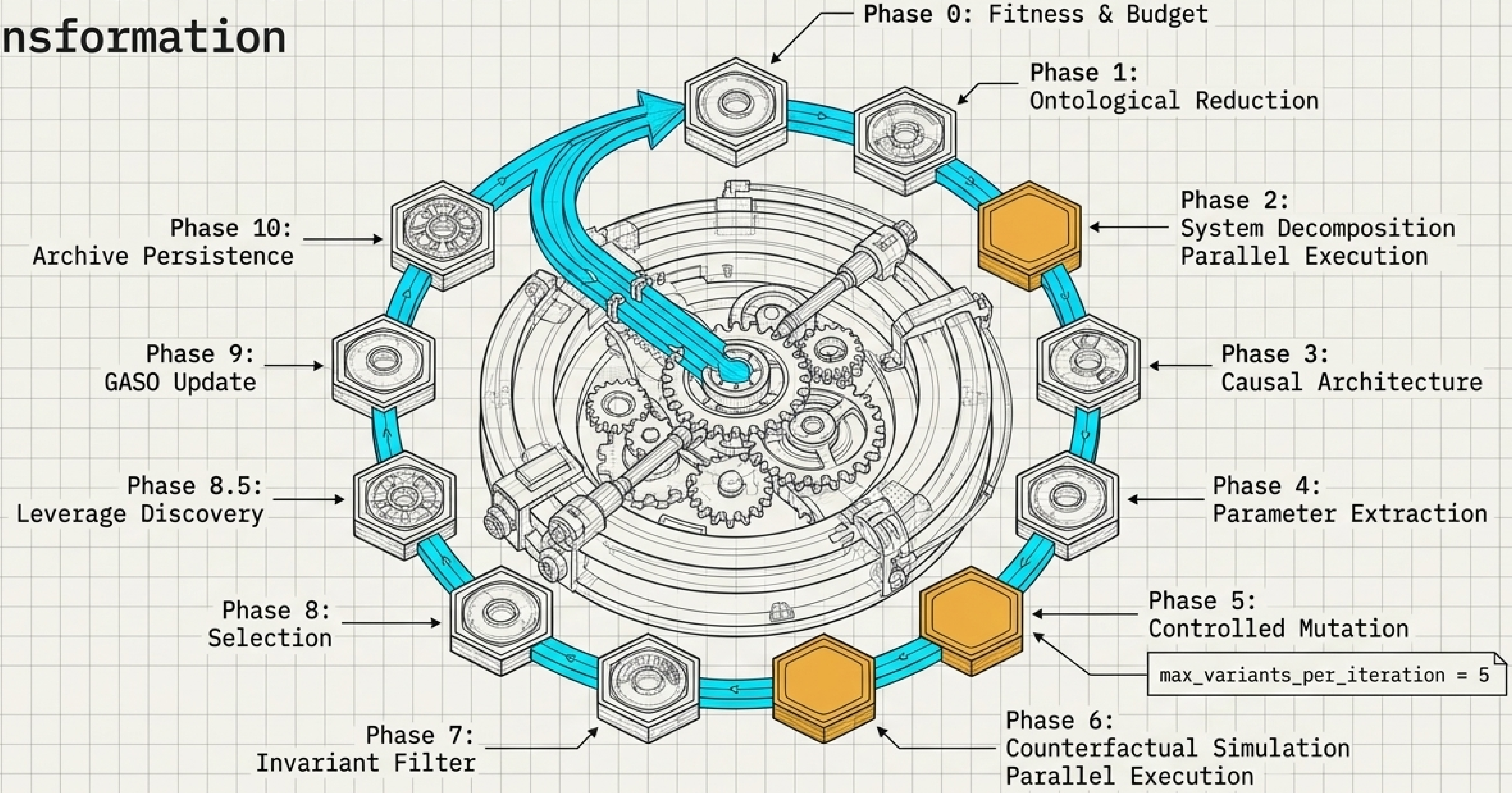


Negative Extraction

The system mandates the archival of failed mutations to permanently map the failure boundaries.



The 11-Phase Evolutionary Engine drives continuous state transformation

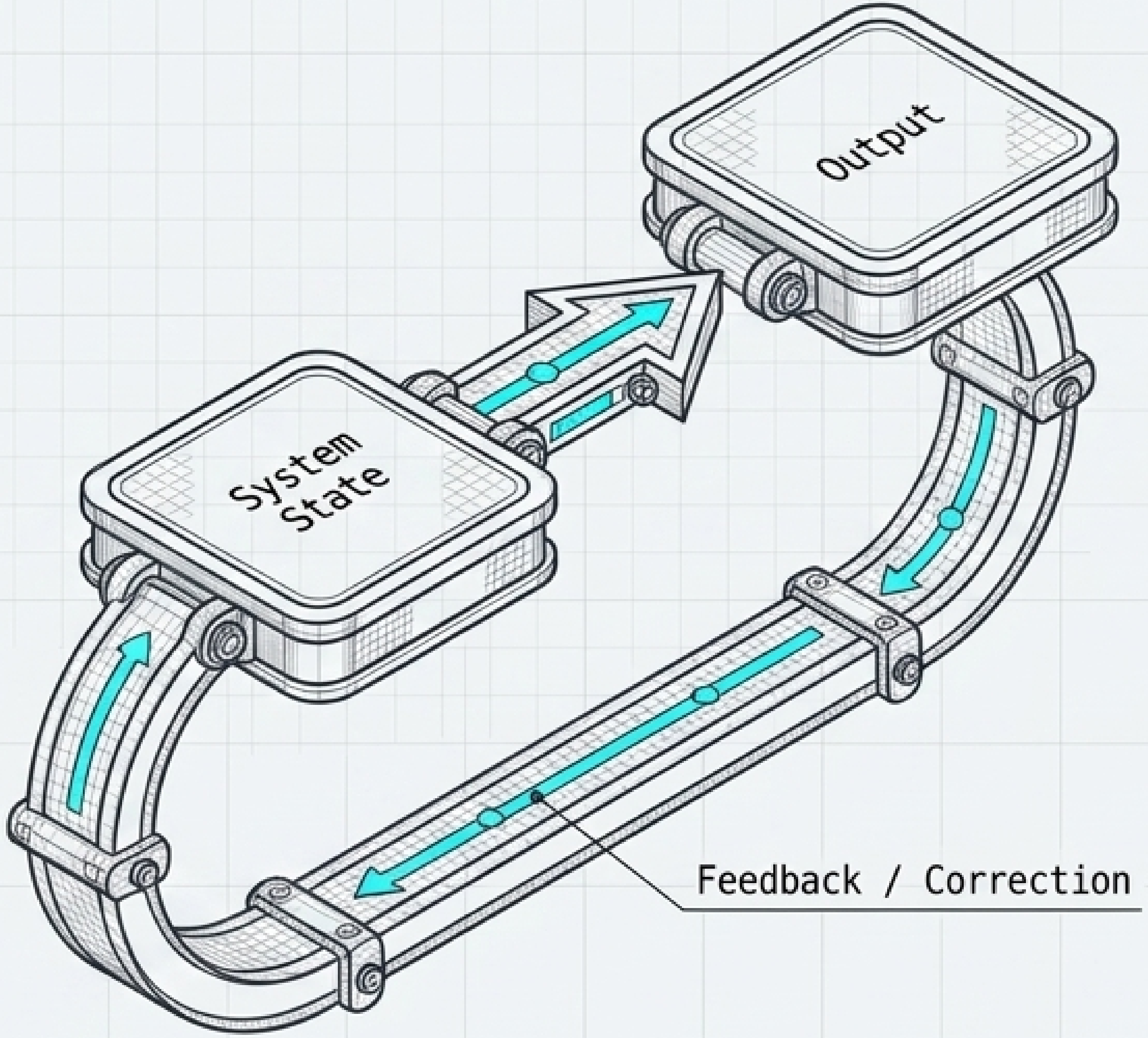


NotebookLM

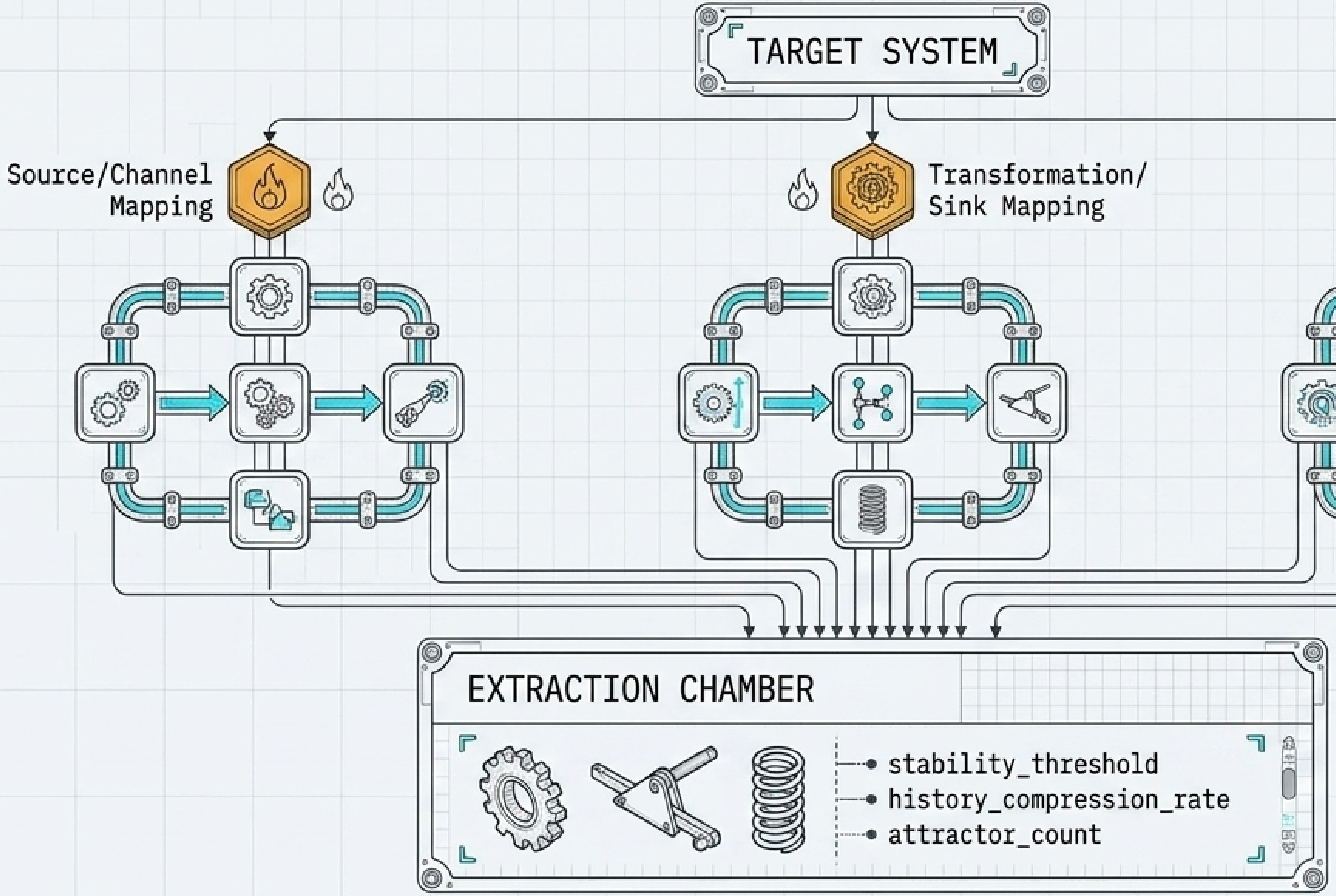
Generation 1 Target:
Wiener's Cybernetics A4 Architecture

TARGET SYSTEM CARD

SYSTEM:	Cybernetic Reasoning Engine (CRE) v2.0
YEAR:	1948, 1950
INITIAL STATE:	predictor_horizon = 0.1s
OBJECTIVE:	Maximize structural regeneration fidelity
RESOURCE ALLOCATION:	Total budget of 15.0 cost units



The engine computationally dissects the architecture to isolate mechanical degrees of freedom



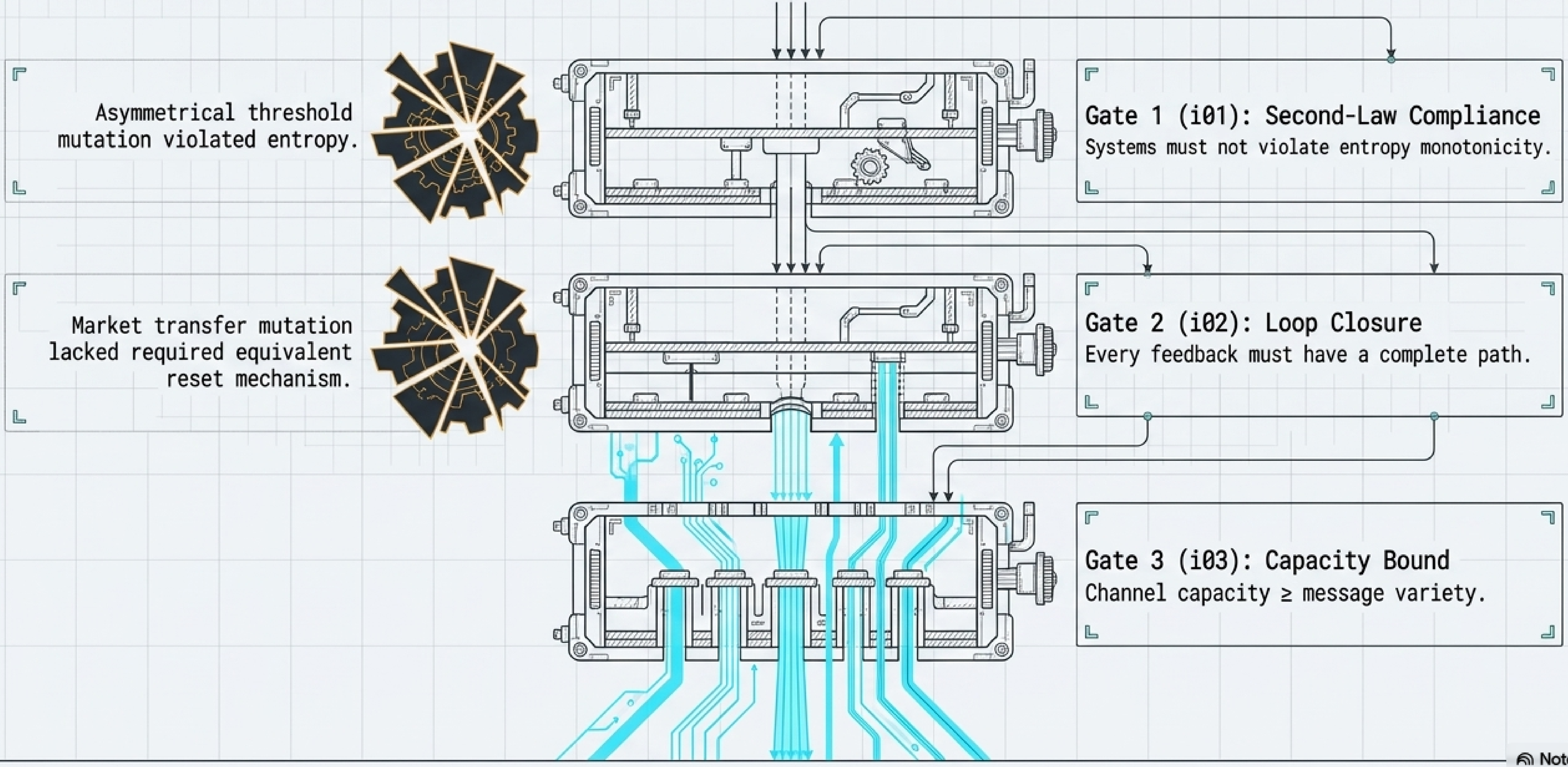
This reduction strips away the narrative domain, leaving only the irreducible causal forces to manipulate.

Five parallel agents executed budgeted mutations targeting specific parameters

TOTAL MUTATION COST: 5.0 UNITS (1.0 per variant)

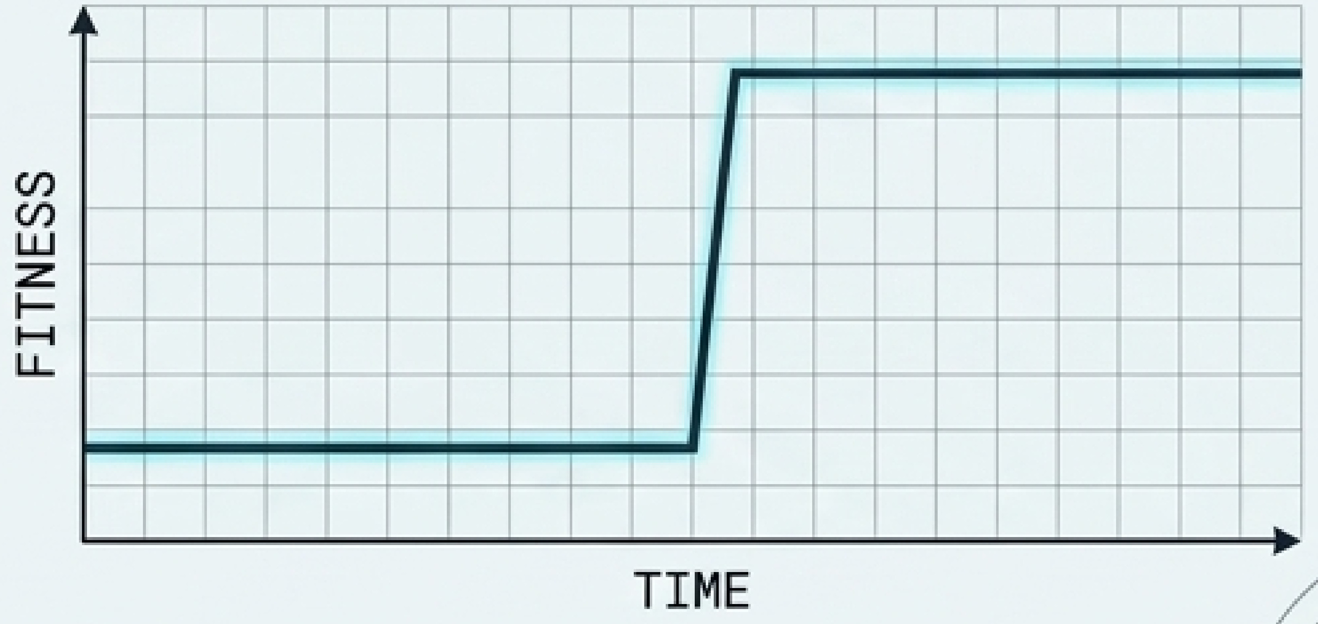
Variant	Operator	Target Variable	Direction	Expected Score
V01	M01 (Quantization)	stability_threshold	Coarser	0.78
V02	M02 (Role-Preserving Transfer)	coord_structure	Neural transfer	0.76
V03	**M03 (Convergence)**	**attractor_count**	**Reduce to 1**	**0.73**
V04	M04 (Disequilibrium)	novelty_injection_rate	Increase	0.65
V05	M05 (Dual-Coordinate)	history_compression_rate	Decrease	0.79

Structural invariants filter unviable mutations into negative constraints



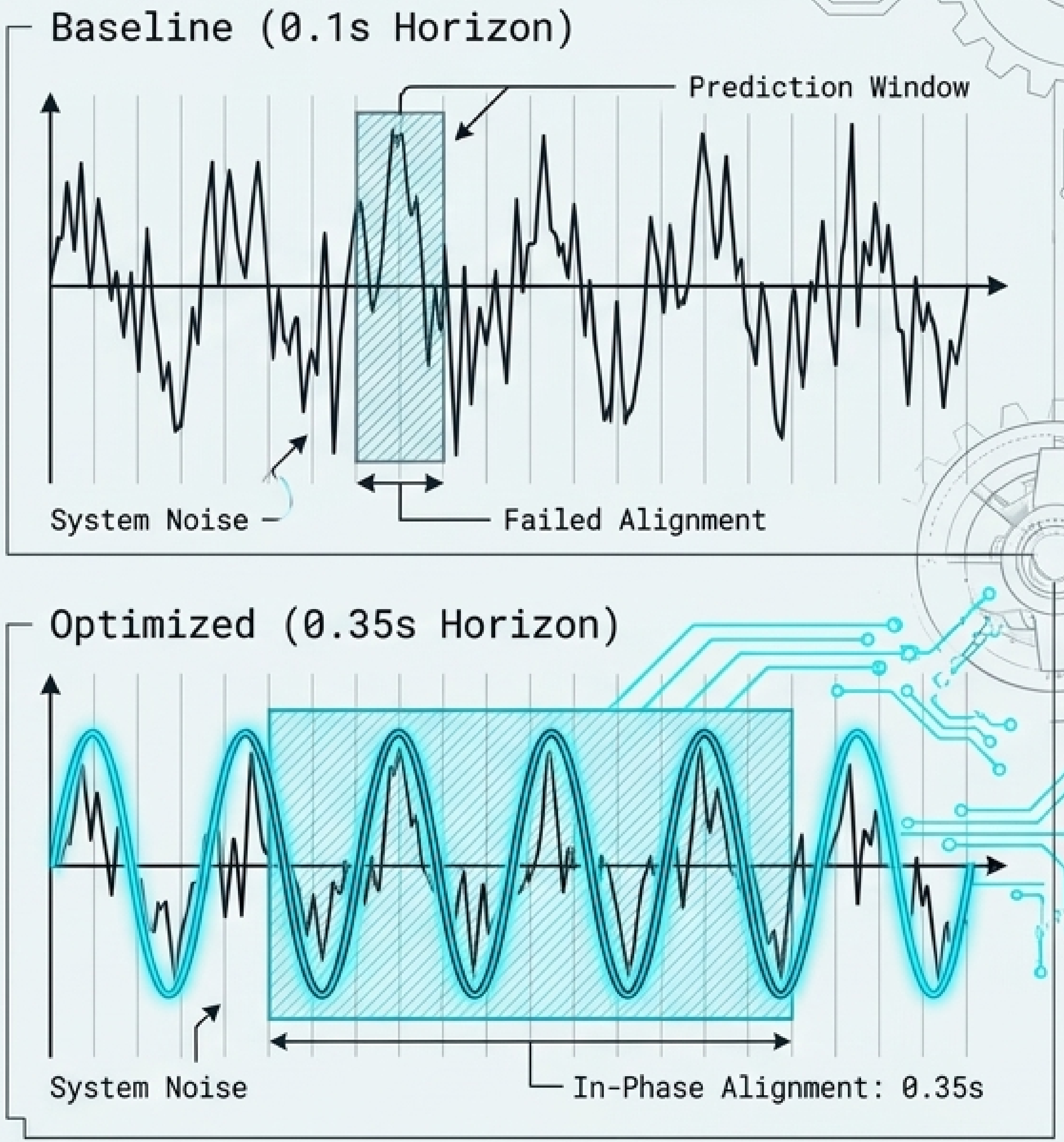
Variant 03 optimizes structural regeneration by matching neural alpha rhythms

WINNER: VARIANT 03



FITNESS INCREASE:
+26% (0.62 -> 0.78)

MECHANICAL CHANGE:
predictor_horizon extended from 0.1s to 0.35s



Maximum regeneration coverage achieved with minimal computational overhead.

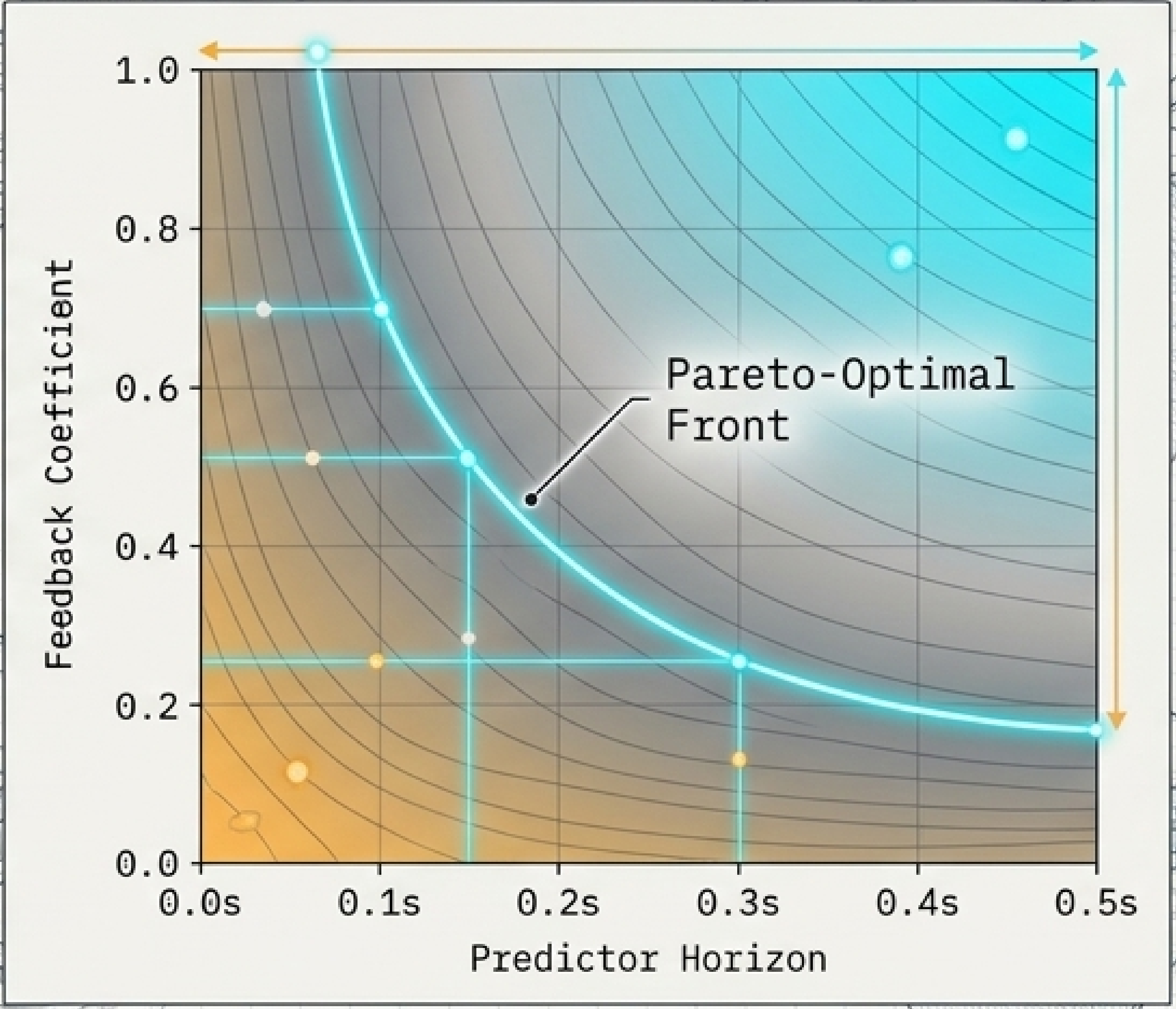
Generation 2 explores the Pareto frontier of foresight versus stability

BUDGET REMAINING: 12.0 / 15.0 cost units (9 iterations)

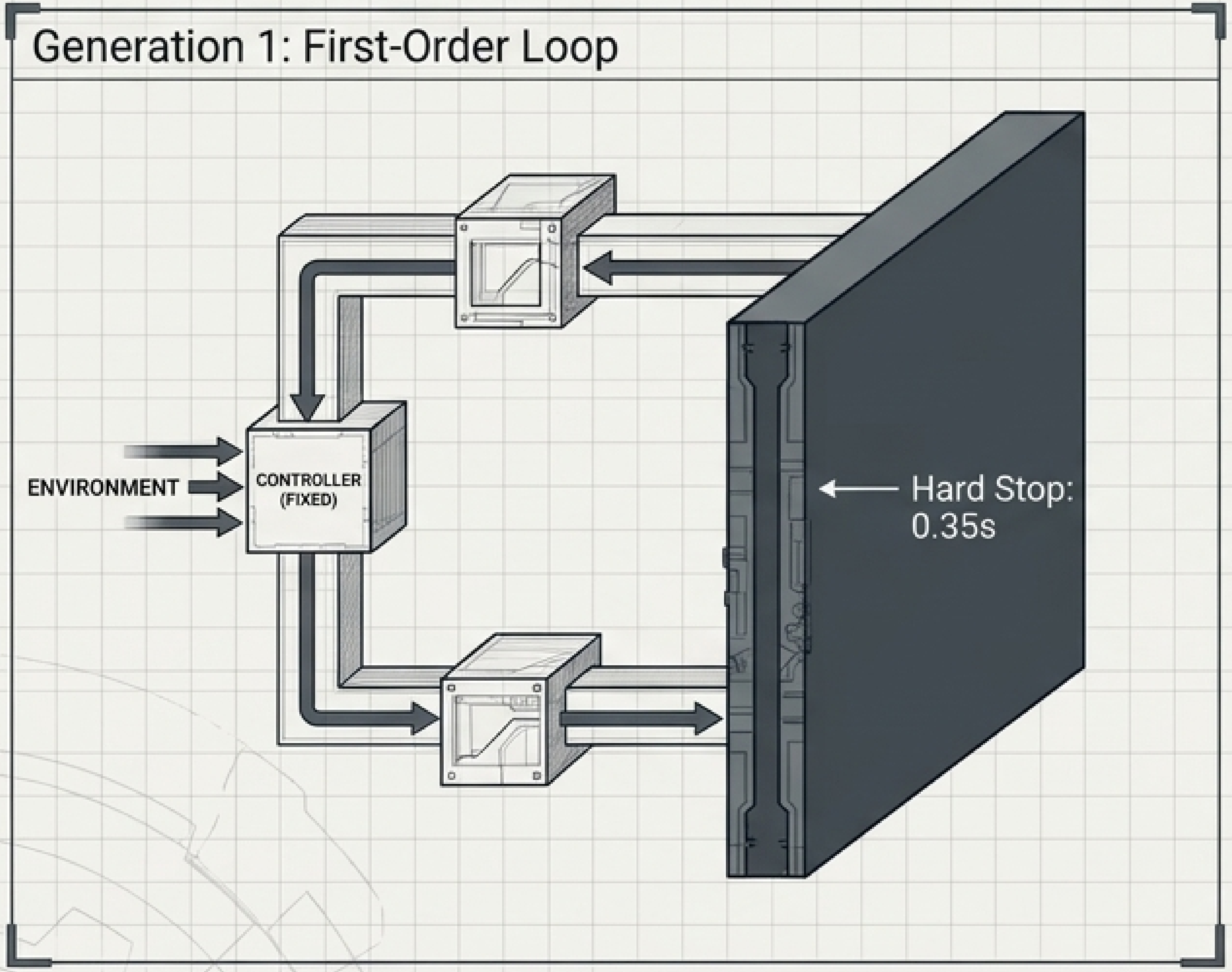
Objective: Optimize the coupled parameter surface to find maximum stabilizing gain for every possible temporal horizon, increasing regime-switch robustness.

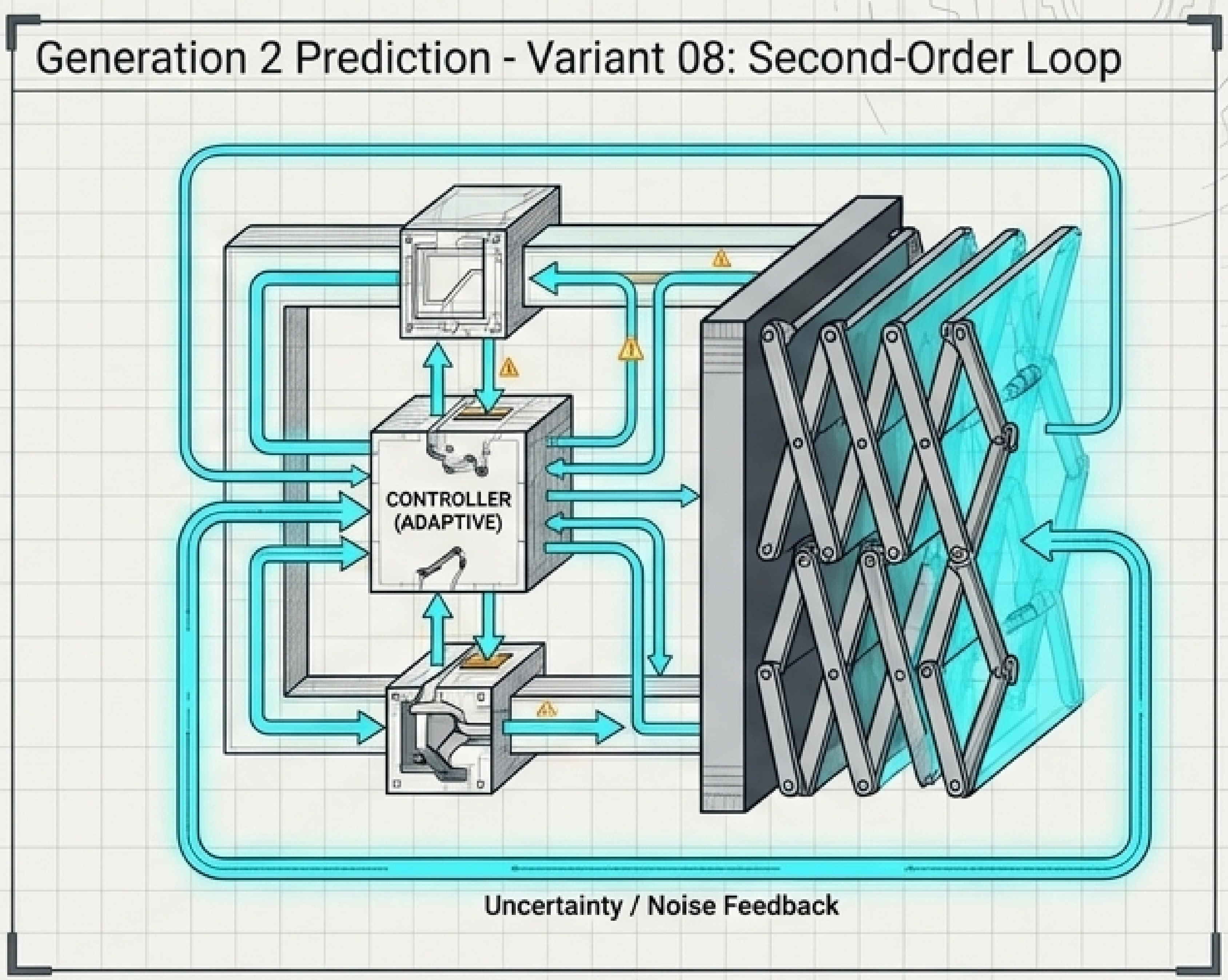
New mutation families deployed:

M06:	Predictor Horizon Sweep
M07:	Feedback Gain Sweep
M08:	Adaptive Horizon Controller
M09:	Predictive Error Weighting
M10:	Channel Capacity Allocation



The anticipated evolutionary leap from fixed foresight to adaptive horizon





The system transitions from merely predicting the future, to dynamically calculating how far into the future it should look based on environmental uncertainty.